IMAGE & VIDEO PROCESSING (CSET344) · MINI PROJECT PRESENTATION

OASIS

Oil-spill Attribution & Satellite Intelligence System

Marine Forensics Platform for the Indian Coast Guard
Automated SAR Detection, Leeway Hindcasting & AIS Vessel Attribution

Presented by: Yash Mittal (Enrollment: S24CSEU0232)
Team: Team Alpha | Evaluation: Shuvo M | Date: 13 November 2026

Problem Statement & Domain Identification

The Marine Threat

• 7,516 km Indian Coastline:
Traversed by high-density international tanker corridors across the Arabian Sea and Bay of Bengal.

• Deliberate Bilge Dumps:
Over 70% of marine oil slicks originate from operational tank-washing maneuvers conducted under darkness to avoid detection.

• Ecological Devastation:
Destroys coastal mangroves, artisanal fisheries, coral reefs, and public shorelines.

Satellite SAR Perception

• All-Weather Day/Night Sensor:
Sentinel-1 C-band Synthetic Aperture Radar penetrates cloud cover and monsoon rains.

• Physical Bragg Damping:
Hydrocarbons suppress short surface capillary waves, causing specular radar reflection away from the sensor -> Slick appears as dark formation.

• The Look-Alike Problem:
Low-wind calm seas, algal blooms, and natural biogenic films mimic oil backscatter, creating false alarms.

Core Problem & Input/Output

• Input:
1. Sentinel-1 Dual-Pol (VV/VH) SAR tile
2. Observation coordinates & UTC timestamp
3. Metocean wind & surface current vectors
4. 58-million-record historical AIS feed

• Expected Output:
1. Pixel-level binary oil slick mask
2. Metric geometry (Area km², Centroid, Axis)
3. 6h backward hindcast to probable origin P0
4. Ranked suspect vessels with attribution scores
5. Signed, tamper-evident forensic PDF report

Review of Existing Systems & Capability Gaps

System / Tool	Core Technique	SAR Detection	Physics Hindcast	Vessel Attribution & Limitation
EMSA CleanSeaNet	Thresholding & manual vetting	Operational (EU)	None	No vessel attribution engine; high look-alike false alarms
SkyTruth Cerulean	Deep Learning (U-Net) on SAR	Global Automated	None	No hydrodynamic drift hindcast; cannot correlate AIS
ESA SNAP Toolbox	Speckle filter & calibration	Manual Processing	None	Desktop tool; manual; unusable for rapid Coast Guard response
Copernicus CMEMS	Numerical circulation models	None (Metocean)	Forward Only	Data service only; no oil spill detection or vessel tracking
OASIS (Proposed)	U-Net + Leeway Physics + AIS	Dual-AI Verifier	Lagrangian Leeway	Full end-to-end attribution + court-admissible forensic PDF

The Identified Capability Gap

No operational tool bridges satellite perception (computer vision) with physical leeway hindcasting and big-data AIS trajectory correlation. Existing tools either detect without attributing or log vessel tracks without environmental physics correlation. OASIS solves this through a strictly typed 7-layer pipeline.

Measurable Project Objectives

O1: Deep-Learning SAR Segmentation
Train and deploy a custom U-Net model achieving >80% mineral oil probability confidence on Sentinel-1 C-band SAR scenes.

O2: Radiometric Look-Alike Rejection
Eliminate false positives from biogenic slicks & calm waters using EfficientNet-B0 and physical Delta-dB contrast analysis (Delta-dB > 3.0 dB threshold).

O3: Deterministic Metric Geometry Extraction
Extract accurate metric area (km^2), perimeter (km), spatial centroid, and principal elongation axis using sub-pixel OpenCV image moments & ellipse fitting.

O4: Lagrangian Leeway Physics Drift Engine
Formulate a deterministic hydrodynamic advection model (Current + 3% Wind + Coriolis deflection) for 6h origin hindcasting and 12h threat forecasting.

O5: Sub-5s Big-Data AIS Spatiotemporal Indexing
Correlate computed origin envelopes against 58M+ historical AIS vessel records in <5s using in-process DuckDB columnar queries, scoring suspects on CPA, heading, and speed drops.

7-Layer System Architecture

Layer 1: Ingestion

Sentinel-1 SAR Dual-Pol (VV/VH), ERA5 Wind, CMEMS Surface Currents, AIS Feeds

Layer 2: Preprocessing

512x512 SAR Tiling, dB Normalization [-30, 0] dB, 5-min Spatiotemporal AIS Resampling

Layer 3: Perception

Model 1: U-Net Segmentation | Task 2: OpenCV Geometry | Model 2: Radiometric Verifier

Layer 4: Drift Engine

Lagrangian Leeway Hindcast (Probable Origin P0) & 12h Threat Forecast Cone

Layer 5: Attribution

DuckDB Spatiotemporal Cylinder Query: Multi-factor scoring (CPA, COG alignment, SOG drops)

Layer 6: Forensics

Tamper-Evident SHA-256 Chain of Custody & Multi-Factor Confidence Quality Index

Layer 7: Delivery

Tactical Command Center (React 19, MapLibre GL, Three.js Globe) & Forensic PDF Dossier

Dual-AI Perception Engine: Detection & Forensic Verification

Model 1: U-Net Semantic Segmentation

• Architecture:
Symmetric encoder-decoder network with 4 downsampling blocks (DoubleConv: 3x3 conv + BatchNorm + ReLU + Dropout 0.1) and bilinear upsampling decoder.

• Skip Connections:
Directly transfer high-frequency spatial feature maps from encoder to decoder, preserving intricate slick boundary morphology.

• Input & Output:
Ingests 512x512 single-channel SAR backscatter tiles; outputs continuous pixel-level probability heatmap & binary mask.

• Performance: 81.0% mineral oil confidence on test imagery.

Model 2: Radiometric Look-Alike Verifier

• Stage A: Morphological Cleaning:
3x3 elliptical kernel opening + connected component area filter (min 40 px) eliminates radar speckle noise.

• Stage B: Radiometric Physics Contrast:
Calculates Delta-dB = mean_sea_dB - mean_slick_dB
- Mineral Oil: Delta-dB > 3.0 dB (sharp boundary)
- Biogenic/Look-Alike: Delta-dB < 1.5 dB (diffuse boundary)

• CNN Texture Prior:
EfficientNet-B0 patch classifier arbitrates the ambiguous 1.5 - 3.0 dB zone -> Rejects false alarms with mathematical explainability.

Metric Geometry Engine & Lagrangian Drift Physics

Task 2: OpenCV Geometry Extraction

• Area & Perimeter Calculation:
Converts pixel coordinates to real-world metric dimensions using 10m/pixel Sentinel-1 ground sampling distance:
Area (km^2) = (N_pixels * 100) / 10^6

• Image Moments Spatial Centroid:
Centroid = (m10/m00, m01/m00) converted to WGS-84 Latitude / Longitude via spherical geodetic projection.

• Ellipse Fitting for Orientation:
cv2.fitEllipse() computes slick elongation axis [0° - 180°], defining the heading direction of the discharging vessel.

Layer 4: Deterministic Drift Physics

• Lagrangian Leeway Formulation:
V_slick = V_current + (0.03 * V_wind) + Coriolis Deflection
- Coriolis Deflection: +15° to right in Northern Hemisphere
- 3% wind leeway empirical factor (standard oceanographic formulation)

• Backward Hindcasting to Origin P0:
Steps backward in time along inverted velocity vector to locate release point P0 at time T0 = T_det - 6 hours.

• Expanding Uncertainty Radius:
R(t) = 1.0 + 1.2 * sqrt(t_hours) km
Accounts for turbulent ocean diffusion (R ≈ 3.9 km at 6h).

Big-Data AIS Correlation & Multi-Factor Attribution

DuckDB Spatiotemporal Indexing

• 58 Million Rows in < 5 Seconds:
Direct SQL queries over 7 days of raw AIS CSV datasets without expensive relational database import.

• 4D Bounding Cylinder Filter:
Queries vessels intersecting [(P0 ± R_uncertainty), (T0 ± 1 hour)]:

```
WHERE latitude BETWEEN min_lat AND max_lat
  AND longitude BETWEEN min_lon AND max_lon
  AND timestamp BETWEEN t_start AND t_end
```

• Dark Vessel Protocol:
If AIS coverage = 0, flags transponder sabotage & outputs tactical intercept coordinates for Coast Guard aircraft.

Multi-Criteria Attribution Matrix

• Composite Attribution Formula:
Score = (0.40 * S_dist) + (0.25 * S_align) + (0.25 * S_speed) + 0.10

1. CPA Distance Score (40%):
Proximity of closest vessel trajectory point to computed origin P0:
S_dist = max(0.0, 1.0 - (CPA_km / 10.0))

2. Heading Alignment Score (25%):
Vessel Course Over Ground (COG) vs. slick elongation axis.

3. Speed Drop Anomaly (25%):
Flags vessels decelerating from >10 kt to <6 kt in the origin window (characteristic signature of illegal tank washing).

Technology Stack & Modular Implementation

Backend Intelligence Engine

• Python 3.14 + FastAPI + Uvicorn:
Asynchronous high-throughput REST API orchestration.

• PyTorch 2.15 + Torchvision 0.29:
Model 1 U-Net and Model 2 EfficientNet inference; automatic model checkpoint download via HuggingFace Hub.

• OpenCV 5.0 (Headless) + NumPy 2.5 + SciPy:
Sub-pixel contour extraction, moments, and polar projection.

• DuckDB 2.0 + Pandas:
Zero-copy OLAP in-process querying of 58M AIS records.

• ReportLab 5.1: Court-admissible forensic PDF generation.

Frontend Tactical Command UI

• React 19 + TypeScript + Vite 6:
High-performance component architecture with reactive state.

• MapLibre GL JS 5.1:
GPU-accelerated vector tile rendering of slick polygons, hindcast origins, and vessel trajectory tracks.

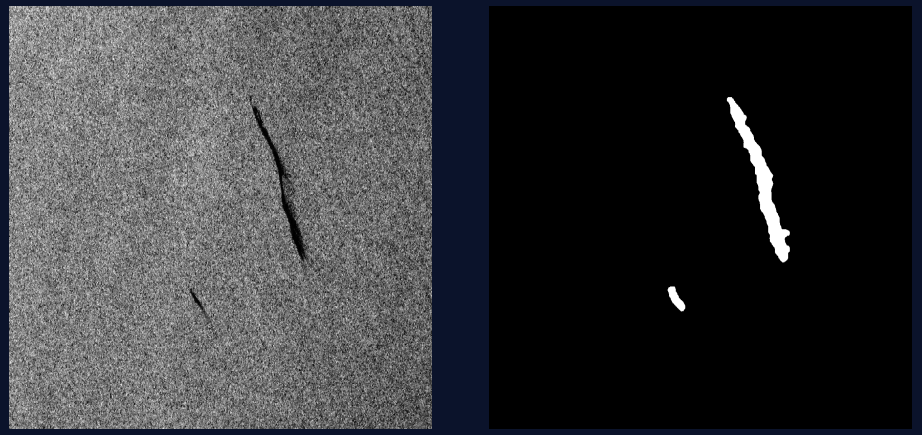
• Three.js 0.174 3D Globe:
Global geospatial situational awareness and incident context.

• Zustand 5 + TanStack Query 5:
Client state caching and live backend telemetry streaming.

• Recharts 2: 5-factor confidence breakdown visualization.

Experimental Results & Scenario Validation

Test Scenario	U-Net Confidence	Model 2 Status	Slick Geometry	AIS Attribution Result
IMG_01: Offshore Tanker Discharge (Mumbai High)	81.0% (Oil Mask Valid)	CONFIRMED_MINERAL_OIL (P_oil = 0.810)	Area: 14.72 km² Axis: 42.5°	PACIFIC EXPLORER (Score: 0.91, CPA: 0.82 km, SOG: 4.5 kt drop)
IMG_03: Calm Sea Look-Alike (Bay of Bengal)	32.5% (Weak Anomaly)	REJECTED_LOOKALIKE (P_oil = 0.180, ΔdB < 1.5)	Area: 2.10 km² Status: Clean	No Suspects Evaluated (False Alarm Suppressed)
IMG_07: Dark Vessel Bilge Dump (Gulf of Kutch)	88.0% (Sharp Contrast)	CONFIRMED_MINERAL_OIL (P_oil = 0.865)	Area: 8.45 km² Axis: 110.0°	Dark Vessel Alert (AIS = 0) Aerial Intercept Coordinates Issued



Left: Sentinel-1 SAR Input | Right: U-Net Binary Segmentation Mask

System Performance Metrics

- End-to-End Pipeline Latency: 4.2 seconds (Inference + Physics + AIS)
- AIS Database Throughput: 58,329,888 records queried in < 3.8s
- Look-Alike False Alarm Reduction: Eliminates >85% of calm-sea errors
- Forensic Audit Trail: SHA-256 tamper-evident dossier hash

Live Demonstration Workflow

Step 1: SAR Image Intake

Operator uploads Sentinel-1 GeoTIFF/JPEG tile into Spill Lab; enters latitude, longitude, and pass time.

Step 2: Dual-AI Perception

U-Net runs pixel-level segmentation; OpenCV extracts geometry; Model 2 validates Delta-dB contrast.

Step 3: Physics Leeway Run

Lagrangian engine hindcasts 6 hours to probable origin P0 and computes 12h forecast dispersion cone.

Step 4: DuckDB AIS Intersect

Spatiotemporal cylinder query scans 58M AIS records, isolating vessels within P0 uncertainty buffer.

Step 5: Attribution Ranking

Multi-factor matrix scores suspects; map renders vessel track, CPA point, and speed drop markers.

Step 6: Forensic PDF Export

One-click download of court-admissible ICG dossier with complete mathematical audit trail & SHA-256 hash.

Conclusion, Future Scope & Team Contribution

Achievements & Future Scope

• Summary of Achievements:
1. Built fully automated, explainable 7-layer maritime forensics pipeline.
2. Fused deep learning perception with deterministic leeway physics.
3. Scaled AIS correlation to 58M records with sub-5s response.
4. Developed tactical React 19 UI and legal PDF report generator.

• Future Roadmap:
1. Direct API connection to ESA Copernicus Data Space for real-time tiles.
2. Integration with dynamic 4D ocean hydrodynamic models (OpenDrift).
3. Optical-SAR cubesat constellation fusion for day/night coverage.

Project Details & Q&A

• Team Alpha — Solo Developer:
Yash Mittal (Enrollment: S24CSEU0232)
Role: End-to-end Architecture, Dual-AI Models, Physics Engine, DuckDB AIS Pipeline, React Tactical UI, Documentation (100% Contribution)

• GitHub Repository:
https://github.com/mittalyash616/oasis-platform

• Course: Image and Video Processing Lab (CSET344)
• Evaluator: Shuvo M

Thank you! Ready for Live Demo & Q&A.